In-Class Assignment L2-1: Running Time Analysis (Group of 2 to 3)

(25 minutes) Submit to LEB2

From the algorithm below,

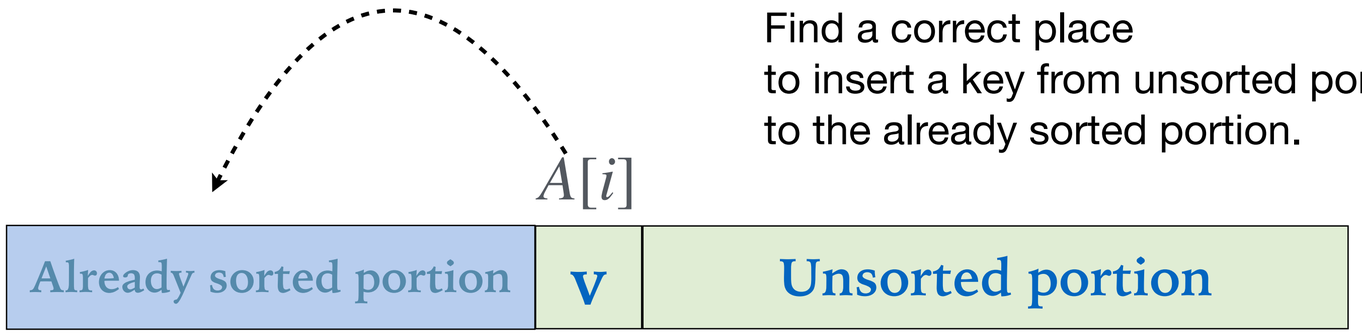
- What task does it solve ?
- What is the basic operation in the algorithm ?
- Are there best-case and worst-case $C(n)$? When do they happen?

  If so, determine $C(n)$ in each scenario.

Algorithm

```
1:  Input: Array A[1 ... n] of integers
2:  Output:
3:
4:  for i = 2 to n do
5:      v ← A[i]
6:      j ← i − 1
7:      while j ≥ 1 and A[j] > v do
8:          A[j + 1] ← A[j]
9:          j ← j − 1
10:     A[j + 1] ← v
```

Solution



Find a correct place
to insert a key from unsorted portion
to the already sorted portion.

Algorithm InsertionSort

```
1:  Input: Array A[1 ... n] of integers
2:  Output: Array A[1 ... n] sorted in a non-decreasing order
3:
4:  for i = 2 to n do
5:      v ← A[i]                          ▷ Current unsorted item
6:      j ← i − 1                         ▷ Starting index in the sorted portion
7:      while j ≥ 1 and A[j] > v do       ▷ Find place to insert v
8:          A[j + 1] ← A[j]               ▷ Shift item to the right
9:          j ← j − 1
10:     A[j + 1] ← v                      ▷ Place key in the proper position
```

Solution

The basic operation is the key comparison $A[j] > v$

Best-case happens when the input is sorted.

- The basic operation in the while loop is only executed once.
- So, $C_{best}(n) = n - 1$

Worst-case happens when the input is reversed-sorted.

- The basic operation in the while loop is executed $i - 1$ times.
- So,

$$C_{\text{worst}}(n) = \sum_{i=2}^{n} \sum_{j=1}^{i-1} 1$$

$$= n(n-1)/2$$